Experiment 6 Day 3: Conversion of Borneol to Camphor

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Introduction

Borneol was converted to camphor through an oxidation reaction using an aqueous sodium hypochloride solution.

Materials and Methods

The experiment began by combining 0.3 g of borneol synthesized in the previous week's reaction with 0.3 mL of glacial acetic acid in a 10 mL round-bottom flask. The racemic mixture was then dissolved in 1 mL of acetone. Next, a distillation column was used to reflux the solution in a warm water bath. Once all reagents were dissolved, 6 mL of aqueous sodium hypochloride was added dropwise over the next 15 minutes while swirling the solution to ensure proper mixing. Next, the mixture was allowed to cool to room temperature and the camphor was extracted using two portions of 2 mL each of methylene chloride (DCM) in a centrifuge tube. The combined methylene chloride layers were then neutralized by adding 2 mL of saturated sodium bicarbonate. Next, the organic solution was washed with 2 mL of 5% sodium bisulfite and 2 mL of water. Finally, the organic solution was dried using anhydrous sodium sulfate and the left-over DCM was evaporated using a warm water bath. The collected product was then weighed to determine yield.

Results

Weight of Watch Glass	37.088 g
Weight of Watch Glass + Actual Yield	37.210 g
Actual Yield (Weight of Product)	0.122 g
Theoretical Yield	0.297 g
Percent Yield	41.08%

Table 1. Actual Yield, Theoretical Yield, and Percent Yield.

Calculations

	Substrate (borneol)	**Product (camphor)**
Amount Used	0.301 g	---
Yield	---	0.122 g
Density	1.011 g/mL	1.218 g/mL
Molar Mass	154.25 g/mol	152.23 g/mol

Table 2. Data Used for Calculations.

Determining theoretical yield.

$mass\ of\ substrate\ used\ = 0.301$ g

$$0.301\ g\ substrate \times \frac{mol\ substrate}{154.25\ g\ substrate} \times \frac{1\ mol\ product}{1\ mol\ substrate} \times \frac{152.23\ g\ product}{1\ mol\ product} = 0.297\ g\ product$$

$$theoretic\ yield\ of\ product = 0.297\ g\ product$$

Determining Percent Yield

$$percent\ yield\ = \frac{actual\ yield}{theoretical\ yield} \times\ 100$$

$$41.08\,\% = \frac{0.122\ g}{0.297\ g} \times 100$$

Discussion/Conclusion

This experiment involved the formation of camphor. The bottom layer in the centrifuge tube was the organic layer which was collected. Given that camphor has a density of 1.218 g/mL compared to water's 1.00 g/mL, water sits on top of camphor as it is much heavier. This implies that the bottom layer was camphor. The experiment's percentage yield was 41.08% reenforcing the notion that there were numerous sources of error leading to a decreased yield. Due to the experiment not being carried out in an anhydrous environment, water could have gotten pulled into the reaction at any point during the experiment allowing impurities to form which

would decrease yield. This could be avoided by placing the reaction tube in a more anhydrous environment where minimal water, if any, would be allowed to enter that vessel. Another source of error could have come from overheating the experiment considering the sensitivity of the reagents to temperature. If the temperature deviated from 40°-50° F, this would have caused superheating leading to evaporation which again, would lead to a decreased yield. This error could be avoided by adding additional boiling chips to the reaction and adjusting the rheostat more often to maintain temperature stability.

References

Keller, Leonard, et al. Organic Chemistry 2 Lab Manual- Auburn University. 2023 ed., Macmillan Learning Curriculum Solutions, 2023